

E-Commerce Sales Dashboard & SQL Analytics

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- Declaration

I now declare that the project work entitled “**E- Commerce Sales Dashboard & SQL Analytics**” is an authentic record of my work carried as part of training in **B.Tech-CSE in ONGC Dehradun.**

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Abstract

The **E-Commerce Sales Dashboard and SQL Analytics Project** was developed to analyse and visualize business performance using a real-world e-commerce dataset. The primary objective of the project was to transform raw transactional data into meaningful insights that support data-driven decision-making. The project involved data preprocessing, SQL-based analysis, data manipulation using Python, and interactive dashboard development in Power BI.

Multiple datasets, including customer information, orders, payments, products, order items, and customer reviews, were integrated and analysed using **Python (Pandas)** and **SQLite**. SQL queries were used to extract valuable business insights such as monthly revenue trends, payment method analysis, top-selling products, seller performance, customer distribution, and order status. Cleaned and summarized datasets were then exported for dashboard creation.

An interactive **Power BI dashboard** was developed to present key business metrics, including Total Revenue, Total Orders, Total Customers, Average Order Value, Monthly Revenue Trend, Payment Method Distribution, Product Category Revenue, Order Status Distribution, and Customer Review Analysis. These visualizations enable users to monitor business performance efficiently and identify important sales patterns and customer behaviour.

The project demonstrates an end-to-end data analytics workflow encompassing data collection, preprocessing, SQL querying, visualization, and business intelligence reporting. It highlights the practical application of Python, SQL, SQLite, and Power BI in solving real-world business problems and generating actionable insights for decision-making.

Acknowledgement

I would like to express my sincere gratitude to Oil and Natural Gas Corporation (ONGC) for providing me with the opportunity to undertake this project as part of my internship. Working on the E-Commerce Sales Analysis Dashboard has been a valuable learning experience that allowed me to strengthen my understanding of data analysis, SQL, Python, and Power BI while applying these technologies to solve real-world business problems. This project has significantly enhanced my analytical thinking, problem-solving abilities, and technical skills in business intelligence and data visualization.

I am especially thankful to my project mentor and the entire ONGC team for their continuous guidance, valuable suggestions, and encouragement throughout the development of this project. Their support helped me understand industry-oriented practices and motivated me to maintain accuracy, consistency, and professionalism in every stage of the work. I also extend my gratitude to my faculty members for providing academic support and motivation during the completion of this project.

Finally, I would like to thank my family and friends for their constant encouragement, patience, and moral support throughout this learning journey. Their motivation inspired me to complete the project successfully. I sincerely hope that this report reflects the knowledge and practical experience gained during the internship and demonstrates the skills developed through this project.

Introduction

The rapid growth of e-commerce has resulted in the generation of massive volumes of transactional data, making data analysis an essential component of business decision-making. Organizations rely on analytical techniques to understand customer purchasing behaviour, evaluate sales performance, optimize product offerings, and improve operational efficiency. By transforming raw business data into meaningful insights, companies can make informed strategic decisions that enhance customer satisfaction and increase profitability.

This project, **E-Commerce Sales Analysis and Interactive Dashboard**, focuses on analysing the Brazilian Olist E-Commerce dataset using **Python, SQL, and Microsoft Power BI**. The project involves the complete data analytics workflow, including data extraction, cleaning, transformation, analysis, visualization, and dashboard development. Python was utilized for data preprocessing and generating summary datasets, SQL was used to perform business-oriented analytical queries, and Power BI was employed to create an interactive dashboard for visual representation of key business metrics.

The dashboard presents important performance indicators such as **Total Revenue, Total Orders, Total Customers, and Average Order Value**, along with analytical visualizations including monthly revenue trends, payment method distribution, order status analysis, product category revenue, and customer review score distribution. These visualizations provide a comprehensive overview of business performance and enable users to identify trends, compare metrics, and derive actionable insights.

The primary objective of this project is to demonstrate the practical application of data analytics techniques in solving real-world business problems. It showcases how raw transactional data can be transformed into meaningful information that supports data-driven decision-making. Furthermore, the project highlights the integration of multiple analytical tools and technologies commonly used in the industry, thereby strengthening practical knowledge in business intelligence, data visualization, SQL querying, and exploratory data analysis.

Overall, this project serves as a comprehensive case study in modern data analytics by combining programming, database querying, and interactive reporting to deliver valuable business insights through an intuitive and professional dashboard.

Objective

The primary objective of this project is to analyse an e-commerce dataset and develop an interactive business intelligence dashboard that provides meaningful insights into sales performance, customer behaviour, payment trends, and product performance. By leveraging **Python, SQL, and Microsoft Power BI**, the project aims to transform raw transactional data into visually appealing and actionable reports that support data-driven decision-making.

To accomplish this objective, the following specific goals were identified:

- To collect, clean, and preprocess e-commerce transactional data for accurate analysis.
- To perform business-oriented data analysis using SQL queries and Python.
- To calculate key business metrics such as **Total Revenue, Total Orders, Total Customers, and Average Order Value**.
- To analyse monthly sales trends, payment methods, order status, customer reviews, and product category performance.
- To design and develop an interactive Power BI dashboard for effective data visualization.
- To present business insights that help understand customer purchasing patterns and overall sales performance.
- To enhance practical knowledge of **Python, SQL, Power BI, data preprocessing, and business intelligence techniques** through the development of a real-world analytics project.

The successful completion of this project demonstrates the practical application of modern data analytics tools and visualization techniques in solving real-world business problems and supporting informed business decisions.

Dataset Description

This project utilizes the **Brazilian Olist E-Commerce Public Dataset**, which contains real-world transactional data collected from an online marketplace. The dataset consists of multiple related tables that provide comprehensive information about customer orders, payments, products, sellers, and customer reviews. These datasets were integrated to perform detailed business analysis and develop an interactive sales dashboard.

The following datasets were used in this project:

Dataset Name	Description
Orders Dataset (olist_orders_dataset.csv)	Contains order-level information such as order ID, customer ID, purchase timestamp, delivery status, and order lifecycle details.
Customers Dataset (olist_customers_dataset.csv)	Includes customer identification and geographic information, such as customer ID, unique customer ID, city, and state.
Order Payments Dataset (olist_order_payments_dataset.csv)	Stores payment-related details including payment type, instalment count, and payment value for each order.
Order Items Dataset (olist_order_items_dataset.csv)	Provides information about products purchased in each order, including product ID, seller ID, item price, and freight charges.
Products Dataset (olist_products_dataset.csv)	Contains product metadata such as product category and other product-related attributes.
Order Reviews Dataset (olist_order_reviews_dataset.csv)	Includes customer review scores and feedback submitted after order completion, which helps evaluate customer satisfaction.

These datasets were merged and processed using **Python (Pandas)** and analysed using **SQL** to generate business insights. The processed data was then imported into **Microsoft Power BI** to create an interactive dashboard that visualizes key performance indicators, sales trends, payment distribution, product category performance, and customer review analysis.

Technologies Used

The successful completion of this project involved the use of various programming languages, database technologies, data analysis libraries, and visualization tools. Each technology played a significant role in different stages of the project, from data preprocessing and analysis to dashboard development and reporting.

Technology	Purpose
Python	Used for data preprocessing, cleaning, merging datasets, feature engineering, KPI calculation, and exporting processed data for visualization.
Pandas	Python library used for efficient data manipulation, transformation, aggregation, and analysis of large datasets.
SQLite	Lightweight relational database used to store the e-commerce dataset and perform SQL-based business analysis.
SQL (Structured Query Language)	Used to retrieve, filter, aggregate, and analyse data through various business-oriented queries such as revenue analysis, customer analysis, and monthly sales trends.
Microsoft Power BI	Used to create an interactive business intelligence dashboard for visualizing KPIs, sales trends, payment methods, product categories, and customer reviews.
CSV Files	Used as an intermediate data storage format for transferring processed datasets from Python to Power BI.
GitHub	Used for version control and project repository management, enabling organized storage and sharing of project files.
Microsoft Word	Used for preparing the project documentation and internship report.

Development Environment

- **Programming Language:** Python 3.x
- **Database:** SQLite
- **Data Analysis Library:** Pandas
- **Visualization Tool:** Microsoft Power BI Desktop

- **Version Control:** GitHub
- **Operating System:** Windows 11

Data Processing

The data processing phase involved transforming raw e-commerce transaction data into a structured and analysis-ready format. Multiple datasets containing information about customers, orders, payments, products, order items, and customer reviews were imported into Python using the **Pandas** library. Each dataset was examined to understand its structure, data types, and relationships with other tables.

Initially, the datasets were inspected for missing values, duplicate records, and inconsistencies to ensure data quality. The relevant datasets were then merged using common keys such as **order_id**, **customer_id**, and **product_id** to create a unified dataset for comprehensive business analysis. Date and time fields were converted into an appropriate datetime format to enable time-based analysis, and a **Year-Month** feature was generated from the order purchase timestamp to analyse monthly revenue trends.

After preprocessing, key performance indicators (KPIs) including **Total Revenue**, **Total Orders**, **Total Customers**, and **Average Order Value** were calculated using Python. Data aggregation techniques such as **grouping**, **sorting**, and **summarization** were applied to generate business-specific datasets for payment methods, order status, monthly revenue, product category revenue, and customer review distribution.

The processed datasets were exported as **CSV files**, which served as the data source for the interactive Power BI dashboard. This structured data processing workflow ensured data accuracy, improved analytical efficiency, and enabled the creation of meaningful visualizations for business decision-making.

Data Processing Workflow

1. Imported multiple e-commerce datasets into Python.
2. Verified data quality by checking for missing values and inconsistencies.
3. Merged related datasets using common identifiers (order_id, customer_id, and product_id).
4. Converted date columns into datetime format and extracted **Year-Month** for trend analysis.
5. Calculated key business metrics such as Total Revenue, Total Orders, Total Customers, and Average Order Value.
6. Generated summarized datasets using grouping and aggregation techniques.
7. Exported processed datasets as CSV files.

8. Imported the processed data into Power BI for dashboard creation and visualization.

SQL Analysis

SQL played a significant role in extracting meaningful business insights from the e-commerce database. The datasets were stored in an **SQLite database**, and various SQL queries were executed to analyse sales performance, customer behaviour, payment trends, and seller performance. Aggregation functions such as **COUNT()**, **SUM()**, **GROUP BY**, **ORDER BY**, and **JOIN** operations were extensively used to retrieve summarized business information from multiple related tables.

The SQL analysis focused on identifying key performance indicators and business trends. Payment transactions were analysed to determine the most frequently used payment methods and their contribution to total revenue. Revenue generated from different order statuses was calculated to understand the financial impact of completed, delivered, and cancelled orders. Monthly revenue trends were obtained by extracting purchase dates and grouping orders based on the year and month of purchase.

In addition, SQL queries were used to identify the top-performing customers based on total spending, top-selling products based on generated revenue, seller performance across different states, and customer distribution across various regions. These analyses helped in understanding purchasing patterns, sales distribution, and overall business performance.

The insights generated through SQL served as the foundation for the Power BI dashboard. The summarized results were further processed using Python and visualized through interactive charts and key performance indicators, enabling efficient business reporting and data-driven decision-making.

SQL Queries Performed

The following analytical SQL queries were executed during the project:

- **Payment Method Analysis** – Identified the number of transactions for each payment method.
- **Revenue by Payment Method** – Calculated total revenue generated through different payment methods.
- **Top Seller States** – Determined the states with the highest number of registered sellers.
- **Revenue by Order Status** – Analysed revenue generated from different order statuses.
- **Top Customers by Spending** – Identified customers with the highest purchase value.
- **Top Products by Revenue** – Calculated revenue generated by individual products.
- **Monthly Revenue Trend** – Evaluated revenue growth over time using monthly sales data.
- **Top Customer States** – Analysed the geographical distribution of customers.
- **Top Sellers by Revenue** – Identified sellers contributing the highest revenue.

The SQL analysis provided valuable business insights that were later integrated into the interactive Power BI dashboard, enabling comprehensive visualization of sales performance and customer behaviour.

Dashboard

An interactive **E-Commerce Sales Dashboard** was developed using **Microsoft Power BI** to present business insights in a clear, visually appealing, and user-friendly manner. The dashboard consolidates key performance indicators (KPIs) and analytical visualizations into a single interface, enabling users to monitor sales performance, customer behaviour, payment trends, and product performance effectively. The processed datasets generated through Python and SQL analysis were imported into Power BI to create dynamic visualizations that support business decision-making.

The dashboard begins with four key performance indicators (KPIs): **Total Revenue**, **Total Orders**, **Total Customers**, and **Average Order Value**, providing a quick overview of the overall business performance. These KPIs help stakeholders evaluate sales growth and customer activity without examining detailed datasets.

Several interactive charts were included to provide detailed analytical insights. A **Monthly Revenue Trend** line chart illustrates changes in revenue over time, helping identify seasonal sales patterns and business growth. A **Payment Method Distribution** pie chart presents the contribution of different payment methods toward overall revenue. The **Order Status** bar chart displays the distribution of completed, delivered, cancelled, and other order statuses, enabling operational performance evaluation. The **Revenue by Product Category** chart highlights the highest revenue-generating product categories, while the **Customer Review Score Distribution** chart provides insights into customer satisfaction by analysing review ratings.

Overall, the dashboard transforms complex transactional data into meaningful visual insights, allowing users to explore business performance efficiently and make informed, data-driven decisions. The interactive design improves accessibility and enables quick analysis of key business metrics, making it a valuable business intelligence tool.

Dashboard Components

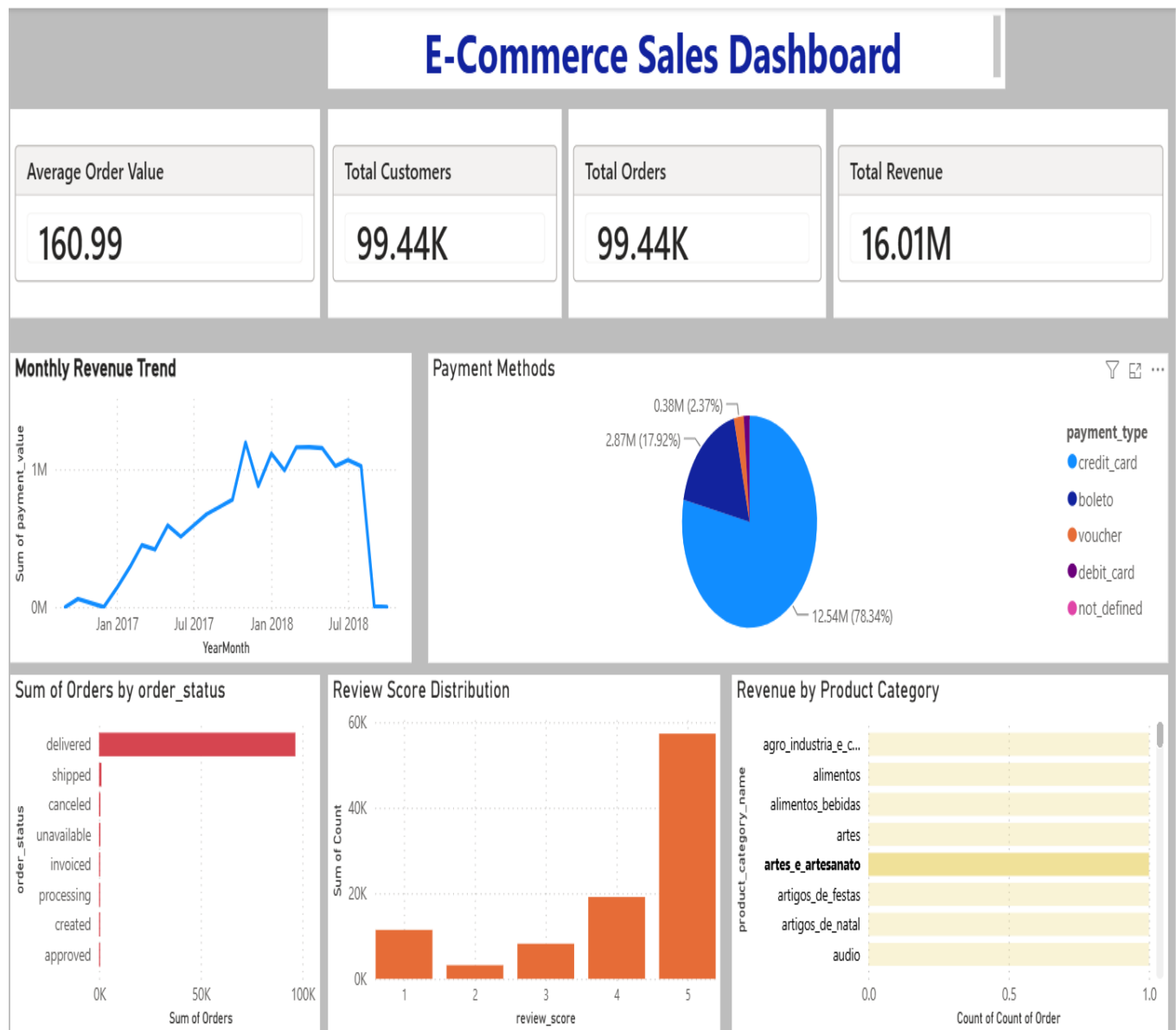
The dashboard includes the following visualizations:

- **KPI Cards**
 - Total Revenue
 - Total Orders
 - Total Customers
 - Average Order Value
- **Monthly Revenue Trend**

- Displays revenue growth and sales trends over time.
- **Payment Method Distribution**
 - Shows the revenue contribution of different payment methods.
- **Order Status Analysis**
 - Illustrates the distribution of orders based on their current status.
- **Revenue by Product Category**
 - Highlights the top-performing product categories based on total revenue.
- **Customer Review Score Distribution**
 - Represents customer satisfaction by displaying the frequency of different review scores.

The dashboard provides a centralized platform for monitoring business performance and supports strategic decision-making through interactive and easy-to-understand visualizations.

Figure 1: Interactive E-Commerce Sales Dashboard



Key Insights

The analysis of the e-commerce dataset revealed several valuable business insights regarding sales performance, customer purchasing behaviour, payment preferences, and product performance.

- **Credit Card** emerged as the most preferred payment method, contributing the highest share of total transactions and revenue. This indicates that customers primarily rely on digital payment methods for online purchases.
- The **Monthly Revenue Trend** showed consistent business growth over most of the observed period, demonstrating stable customer demand and increasing sales performance, with slight fluctuations during certain months.
- Analysis of **Order Status** revealed that the majority of orders were successfully delivered, indicating efficient order fulfilment and a relatively low percentage of cancelled or unavailable orders.
- The **Product Category Revenue Analysis** identified a small number of product categories that generated a significant portion of the overall revenue. This information can assist businesses in inventory planning, promotional campaigns, and product portfolio optimization.
- Customer review analysis showed that **4-star and 5-star ratings** accounted for the majority of reviews, reflecting a high level of customer satisfaction with products and services. However, lower ratings suggest opportunities for improvement in product quality, delivery time, and customer service.
- The dashboard highlighted that **Average Order Value (AOV)** remained relatively stable, providing an important performance metric for evaluating customer purchasing behaviour and revenue generation.
- SQL-based analysis also identified **top-performing customers, products, and sellers**, enabling businesses to recognize valuable customers, high-revenue products, and successful sellers for future business strategies.

Overall, the project successfully transformed raw transactional data into actionable business insights through SQL analysis, Python-based data processing, and interactive Power BI visualizations. These insights can support strategic decision-making, improve operational efficiency, enhance customer satisfaction, and contribute to overall business growth.

Challenges

During the development of this project, several technical and analytical challenges were encountered while working with the large-scale e-commerce dataset. One of the primary challenges was integrating multiple datasets containing orders, customers, payments, products, order items, and reviews. Establishing relationships between these datasets using common identifiers while maintaining data consistency required careful preprocessing and validation.

Another significant challenge was performing complex SQL analysis to extract meaningful business insights. Designing optimized SQL queries involving multiple **JOIN**, **GROUP BY**, **ORDER BY**, and aggregate functions required a thorough understanding of database relationships and query optimization techniques. Ensuring the accuracy of calculated metrics such as revenue, customer spending, and monthly sales trends was also an important aspect of the analytical process.

Developing the interactive dashboard in **Microsoft Power BI** presented additional challenges, particularly while importing processed datasets, configuring data relationships, resolving data refresh issues, selecting appropriate visualizations, and designing an intuitive dashboard layout. Considerable effort was invested in ensuring that the dashboard accurately represented business metrics and provided meaningful insights in a clear and user-friendly format.

Despite these challenges, systematic debugging, continuous testing, and iterative improvements helped overcome the issues successfully. These experiences significantly enhanced practical knowledge of **Python, SQL, data preprocessing, business intelligence, and dashboard development**, while strengthening analytical thinking and problem-solving skills required for real-world data analytics projects.

Project Highlights

Unlike a basic data visualization project that focuses only on displaying charts and reports, this project follows a complete **end-to-end data analytics workflow**. It begins with raw e-commerce datasets and progresses through data preprocessing, SQL-based business analysis, feature engineering, dashboard-ready data generation, and interactive visualization using Microsoft Power BI. This structured approach demonstrates practical implementation of the complete data analytics lifecycle rather than simply presenting graphical reports.

A key strength of this project is the integration of multiple technologies, including **Python, SQL, SQLite, and Power BI**, to solve a real-world business problem. Python was used for data cleaning, preprocessing, feature engineering, KPI generation, and exporting dashboard-ready datasets. SQL was utilized to perform business-oriented analytical queries and generate valuable insights, while Power BI transformed the processed data into an interactive business intelligence dashboard.

The project also focuses on generating actionable business insights instead of merely presenting raw statistics. Business metrics such as **Total Revenue, Average Order Value, Monthly Revenue Trend, Payment Method Distribution, Order Status Analysis, Product Category Revenue, and Customer Review Analysis** enable stakeholders to understand business performance, customer behaviour, and sales trends more effectively. This analytical approach makes the dashboard suitable for supporting real business decision-making.

Overall, the project demonstrates practical skills in **data preprocessing, SQL querying, business intelligence, dashboard development, and data-driven decision-making**, making it more comprehensive than a conventional reporting project that only displays charts without meaningful analytical interpretation.

Conclusion

The **E-Commerce Sales Analysis and Interactive Dashboard** project successfully demonstrated the complete process of transforming raw transactional data into meaningful business insights through data analytics and business intelligence techniques. By integrating Python, SQL, SQLite, and Microsoft Power BI, the project covered every stage of the analytics pipeline, including data collection, preprocessing, analysis, visualization, and reporting.

The developed dashboard provides a comprehensive overview of business performance through key performance indicators, revenue trends, payment analysis, product category performance, order status distribution, and customer review analysis. These visualizations enable stakeholders to monitor important business metrics efficiently and support informed, data-driven decision-making.

In addition to strengthening technical skills in Python, SQL, and Power BI, this project enhanced my understanding of data preprocessing, business analysis, dashboard design, and analytical thinking. The experience gained during the project has provided valuable practical exposure to real-world data analytics workflows and business intelligence practices, making it a significant learning experience during my internship.

The project successfully achieved its objectives and demonstrates how modern analytical tools can be integrated to convert complex business data into actionable insights that contribute to improved decision-making and organizational performance.

References

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